Neeraj Vyas

Data Engineer

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Ambitious Data Engineer with strong knowledge in Data Pipelines, ETL.

Expertise in Apache Airflow, Spark (Moderate), AWS, Python.

Certified in both Apache Airflow certifications from Astronomers.

Skills

- Apache Airflow On Prem and on Cloud as well (AWS MWAA, Astronomers
- Apache Spark, Hive, Hadoop, Oozie
- AWS - MWAA, EKS, ECS, ECR, EMR, IAM, EC2, Data-Pipelines, Lambda, RDS, S3. Python and Shell scripting, (Unix - Linux and Mac)
- SQL and NoSQL (Mysql, RDS, Redshift, MongoDB and Elasticsearch
- Git, Docker, Kubernetes and Helm
- AWS Lambda , Cloudwatch, Push (SNS), Pull (SQS)

I have been selected for doing M Tech. in Quantum Computing from SGSITS, Indore, M.P. (July 2026) Might join in future

Work History

2024-04 - 2025-06

Team Lead in BancApp Automations

- Took multiple interviews of Data Engineers and interns.
- Started RnD on other services which are similar to AWS
- Presentation skills, Give 2 Technical Presentation to whole technical team
- NETC - RnD Fast tag on toll plaza car tuck images
  - object, timestamp, number plate of car/truck
- RnD on Pandas, Dask
- Last time did some work related to K8s.

2024-04 - 2025-05

RnD Data Software Engineer in Banc App Automations

- Did understanding on streaming analysis, it's use-cases.
- Did RnD on Apache Kafka, Deployed it on Azure VM
- Kafka deployment in 3 ways
  - docker
  - zookeeper
- AWS Glue ETL Functions
- Rnd Amazon EMR.
- AWS Lambda and AWS Cloudwatch
- Durable Functions extension for stateful workflows
- Gave the demo to whole company.
- PDF Parser, Adaptor for the product of the company.
  - & many more

2022-04 - 2022-08

Certified in **Apache Airflow** which is an essential part of Data Engineering and Medical Issue, **(Career Break)**

Keystone Strategy, USA, Remote from Indore
Collaborated on ETL (Extract, Transform, Load) tasks, maintaining data integrity and verifying pipeline stability.

Multiple AWS services setup and access management

- MWAA v2, Dev and Prod.
- S3 and IAM
- RDS Setups based on requests.

Migration of Data pipelines

- Migration of all the data pipelines from airflow's v1 to v2.
- Verification and maintenance of pipelines and MWAA.

New Data Pipeline

- Pipeline to get data from Diffbot.
- Firmographic data of companies.
- Storage in mongo, Redshift and S3 based on requirement.
- Outlook Email HTML reports and Slack Alerts.

Debugging of all AWS services

- Pipeline maintenance
- Streamlit webserver issue.
- Others requested by data scientists.

Data requests by Data scientists.

- Management of RDS and Redshift.
- Data preprocessing and Analysis on Sagemaker.

2021-07 - 2022-04

Data Engineer

Keystone Strategy, USA, Remote from Indore, India

Multiple Setups and management on AWS related to Data Engineering

- Airflow Setup on EC2, Sequential and Local Executor.
- MWAA Setup v1 and v2.
- RDS Setup (Multiple based on data scientists)
- Data Work on AWS Sagemaker, Redshift and Mongo DB.

Data pipelines - Creation and Management Majority tasks were

Web Scrapping,
Storing Raw Data in Mongo,
Data Preprocessing,
Creation of EMAIL Reports and send on Outlook,
Slack success and failure Alerts.

Below were some Data Pipelines - Scheduled or Manually Triggered.

- DOJ news pipeline (Department of Justice)
- https://www.justice.gov/news USPTO Patent Processor
- https://www.uspto.gov/patents OCR Tool - Docker Image creation, AWS ECR and ECS usage.
- Papermill - Execution of Jupyter Notebooks.
- Multiple Prospector Data Pipelines (https://www.sec.gov)
- Court Cases, (Example Amazon vs Facebook)
- Private equity Sell-Side
- Security digital innovation

Multiple Data Requests by Data Scientists (Work on AWS Sagemaker)

- Data Preprocessing and Analysis (Requirement of specific data from Redshift Data, {Date-Ranges, Sorted, Fortune 100 and so on...})
- Data Loading From S3 - Redshift - Mongo - RDS and multiple vice verses.
- Data Visualization (Streamlit)
- Creation of multiple RDS and Resource management.

2020-11 - 2021-07

Data Engineer

American Freelancing Platform, US

- **Apache Airflow, K8s, Helm, Git Sync** - Deployed Apache Airflow on Digital Ocean's k8s cluster. **Airflow Dags development-** Designed dags
- to test Airflow Environment.

2020-07 - 2021-07

Software Engineer

Impetus, India

- **Apache Airflow -** Configured multiple Airflow Setups - on EKS (K8s Executor - Docker, k8s, helm), - on EC2 (Celery Executor), Conversion of complex Apache Oozie workflows into Airflow DAGs. Created scheduled and On Demand Data-Pipelines in Airflow, Integration of Apache Livy, Hive, Spark jobs in Airflow. POCs on AWS MWAA. (Did Certification from Astronomer for Apache Airflow Fundamentals) **Spark on EKS -** POCs on GCPs **spark-k8s-operator**, Created spark jobs to be executed on EKS
- from **Airflow** with use of Hive-Server2, From this approach client is not dependent on AWS EMR, resulting in huge cost saving. **ETL Optimizations -** Used **Apache Hive** and **Apache Spark** and optimized complex **Apache Oozie** workflow running on **AWS EMR** Cluster,
- previously It was taking 23~24 hours, after optimization it gets completed in 10 hrs for each workflows. (Got star of month for above work.) **Airflow, Spark, Hive, Bash, Python** - Created Partition Compaction pipeline for compacting partitions of hive from Spark using Airflow.
- Results in Faster query on data-set. **AWS Data-Pipeline**, **Lambda, EMR, DynamoDB, Elastic Search, S3** - Cluster split approach for loading data into Elastic Search Cluster from hive connector tables resulted in cost
- savings and added process lineage to track file dates for ETL workflows. (Got star of month for Cluster split approach work.)

2019-05 - 2020-06

Associate Software Engineer

Impetus, India

- **Kafka** - Created S3 connectors which places data on S3 in parquet format from kafka topics and did analysis on them by creating hive tables on S3. **Open Source Patches** - Elastic Search upsert support in
- ES-Hadoop jar, Added Hive and Oozie patches during EMR version upgradation. **PoC on Apache Airflow**, started from scratch, Basic
- setups on Local Box with mysql metadata. **AWS Lambda** - Add Partition Function, kafka streamer places files on S3 (Hive warehouse), to sync
- with hive metastore created this lambda

function to add partitions in hive table.

- Debugged code and located root causes of problems by reviewing configuration files and logs.
- **Case Study on Risk Management system** (microservices) - Spring Boot, Angular

2018-12 - 2019-02

Internship Student

Systango

- Web development using Javascript and
- Bootstrap. Hosted Web application on Heroku (https://headphonestore24.herokuapp.com/)

Education

2015-06 - 2019-05

Bachelor of Engineering: Computer Science

Swami Vivekanand College of Engineering

- Scored 8.52 CGPA on 10 Point scale.

Accomplishments

- Bright Beginner Award in Impetus
- 3 Star of the Months Award in Impetus
- Certificate by “National Programming Aptitude Test (NPAT)” for scoring 81 percentile in the programming test organized by NPTEL in collaboration with ACM India in 2017.
- 3 Hackathons in different cities, (Created 3 different projects as Team Leader in each, which were based on Desktop App, Web App and IoT)

1. Certificate of Participation in Finale at World's Biggest "Smart India Hackathon, 2017” organized by Indian Government.
2. Certificate of Participation at “SRM Hackathon, 2017” organized by SRM University.
3. Certificate of Participation at IOT based “Bhopal Smart City Hackathon, 2018” organized by Bhopal government and Hewlett Packard Enterprise.

Certifications

2023-02 Astronomer Certification DAG Authoring for Apache Airflow

2021-03 Astronomer Certification for Apache Airflow Fundamentals

2018-02 Basics of Data Science in R From NPTEL IIT Madras, India